INCH-POUND

MIL-DTL-17361/11B(SH)
19 September 2006

SUPERSEDING
MIL-C-17361/11A(SH)
13 March 1995

DETAIL SPECIFICATION SHEET

CIRCUIT BREAKERS, AIR, ELECTRIC, INSULATED HOUSING (SHIPBOARD USE) TYPES AQB-LF400 AND AQB-LF402

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17361.

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TABLE I. Requirements.

Maximum rated continuous current	400 amperes
Voltage and frequency	500 VAC, 60 Hz 250 VDC (AQB-LF400 only)
Interrupting rating	100,000 amperes, 60 Hz, AC
Peak let-through current	45,000 amperes
Trip unit	Interchangeable
Current limiting fuses	LF5H25ORD, 14-000–20,000 amperes LF5H40ORD, 17,000–30,000 amperes
Type of connections	Back (removable mtg.)
Electric operator	Available
Undervoltage trip or shunt trip	Either available
Auxiliary switch	Two (a) contacts; two (b) contacts
Handle lock	Group 3 (Figure 1 of MIL-DTL-17361)
Handle boot	Not available

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TABLE II. Continuous current ratings settings.

AQB-LF400 trip units only							
Element rating /1							
250T	250TM	300T	300TM	350T	350TM	400T	400TM
AQB-LF402 trip units only							
Continuous current setting (amperes)							
250		300		350		400	

NOTE:

/1 Identification marking of trip units:

Numerals indicate continuous current rating of the specific trip unit.
Letter “M” denotes calibration for applications having a high inrush current.
Letter “T” denotes that the trip units are available to AQB-LF400.

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TABLE III. Adjustable instantaneous trip settings amperes/maximum time (all 3 poles).

AQB-LF400 trip units only /1												
Element rating	**LO /3**		**2**		**3**		**4**		**HI /3**		**Seconds /2**	
	AC	**DC**	**AC**	**DC**	**AC**	**DC**	**AC**	**DC**	**AC**	**DC**	**60 Hz AC**	**DC**
250T	880	---	1035	---	1165	---	1325	---	1530	---	0.024	0.028
250TM	3190	---	3725	---	4325	---	5175	---	5825	---		
300T	1380	---	1610	---	1820	---	2035	---	2280	---		
300TM	3190	---	3725	---	4325	---	5175	---	5825	---		
350T	1380	---	1610	---	1820	---	2035	---	2280	---		
350TM	3190	---	3725	---	4325	---	5175	---	5825	---		
400T	1380	---	1610	---	1820	---	2035	---	2280	---		
400TM	3190	---	3725	---	4325	---	5175	---	5825	---		

AQB-LF402 trip units only /1 /4										
Frame rating	**3X**	**4X**	**6X**	**8X**	**10X**	**12X**	**15X**	**20X**	**30X**	**40X**
400	1200	1600	2400	3200	4000	4800	6000	8000	12000	16000

NOTES:

/1 The tolerance for each ampere setting shall not exceed ±10%.

/2 Time shall be measured at three times pickup current of instantaneous setting. For qualification tests only, not for routine calibration.

/3 The LO and HI settings only are calibrated. Others are approximate.

/4 Production units shall be calibrated at the 8X setting. Opening time shall be 0.025 seconds or less when tested at the 3X instantaneous pickup (for qualification tests only, not for routine calibration).

TABLE IV. Adjustable short-time delay pickup (STDPU) settings in amperes (STD) (all 3 poles) (for trip unit types AQB-LF402 only).

Continuous current setting	**2X**	**3X**	**4X**	**6X**	**8X**	**10X**	**15X**
250	500	750	1000	1500	2000	2500	3750
300	600	900	1200	1800	2400	3000	4500
350	700	1050	1400	2100	2800	3500	5250
400	800	1200	1600	2400	3200	4000	6000

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TABLE V. Adjustable short-time delay (STD) settings in seconds (for trip unit types AQB-LF402 only).

STD "delay setting"		
1	**2**	**3**
0.025–0.055 /1	0.065–0.095	0.105–0.135

NOTE:

/1 Time delay with 2X STDPU settings shall be 0.020–0.060 seconds.

TABLE VI. Tripping characteristics – long-time delay (LTD) (all 3 poles).

AQB-LF400 trip units only	
Current as multiple of trip element rating	**Tripping time 60-Hz AC trip element DC trip element**
1.30	Not less than 1 hour
2.25	Less than 1 hour
6.00	Between 20 and 32 seconds
AQB-LF402 trip units only	
Current as multiple of CCS	**Tripping time 60 Hz and 400 Hz AC**
1.35	Not less than 1 hour
1.65	Less than 1 hour
4.0	Between 35 and 55 seconds
6.0	Between 20 and 32 seconds

TABLE VII. Interruption performance test requirements /1.

Circuit sample	Element amperes	Interrupting amperes	Power factor lagging	Unit operation	See note
1A	400	100,000	0.15–0.20	O-CO-CO	/2
1A(1)	400	17,000	0.15–0.20	O-CO-CO	/3 /4
1A(1)		30,000	0.15–0.20	O-CO-CO	/2 /3
1A(1)		75,000	0.15–0.20	O-CO-CO	/2 /3
1B	250	100,000	0.15–0.20	O-CO-CO	/2

NOTES:

/1 Oscillograms shall be taken on all interrupting tests.

/2 Circuit breaker shall open and at least one fuse blow.

/3 New circuit breakers may be used for tests above the standard circuit breaker rating.

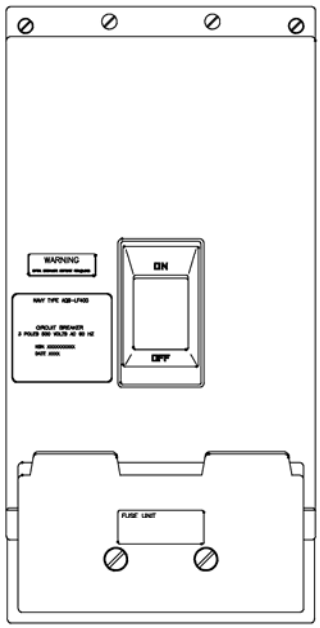
/4 Circuit breaker shall open on instantaneous trip and fuses shall not blow.

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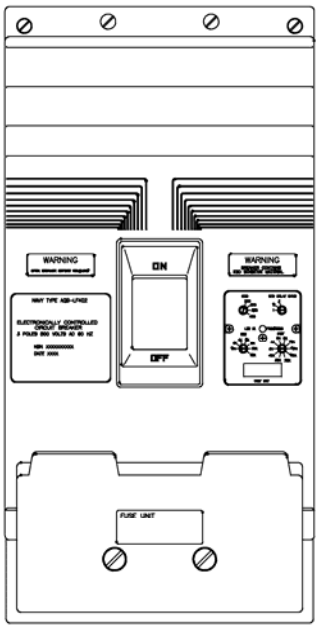
TABLE VIII. Endurance and temperature rise test requirements.

<table>
<tr><td>Electrical endurance test</td><td colspan="4">The rate of operation shall be 5 operations per minute for a total of 4000 close-open operations.</td></tr>
<tr><td rowspan="4">Temperature rise tests</td><td colspan="4">Size of cable and bus bars for temperature rise tests</td></tr>
<tr><td rowspan="2">Copper rating-amperes</td><td colspan="2">Cable size</td><td>Copper bus size (inch)</td></tr>
<tr><td>AC</td><td>DC</td><td>AC and DC</td></tr>
<tr><td>400</td><td>Two (2) AWG 000</td><td>400 MCM</td><td>1-1/2 x 1/8</td></tr>
</table>

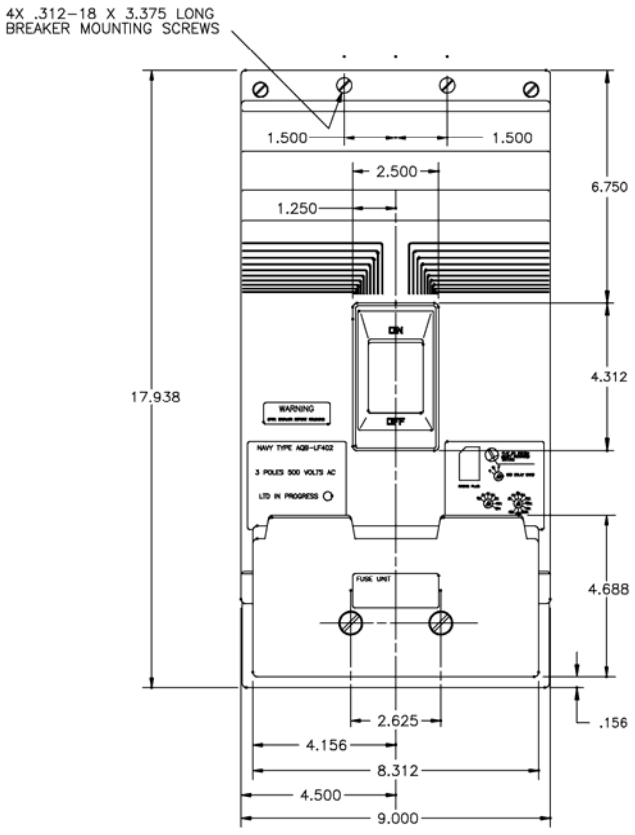
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FRONT VIEW OF AQB–LF400 type CIRCUIT BREAKER



FRONT VIEW OF AQB–LF402 type CIRCUIT BREAKER
WITH A CCS SWITCH



FRONT VIEW OF AQB–LF402 type CIRCUIT BREAKER
WITH RATING PLUG

DIMENSIONS TYPICAL

FIGURE 1. Circuit breaker types AQB-LF400 and AQB-LF402 outline dimensions.

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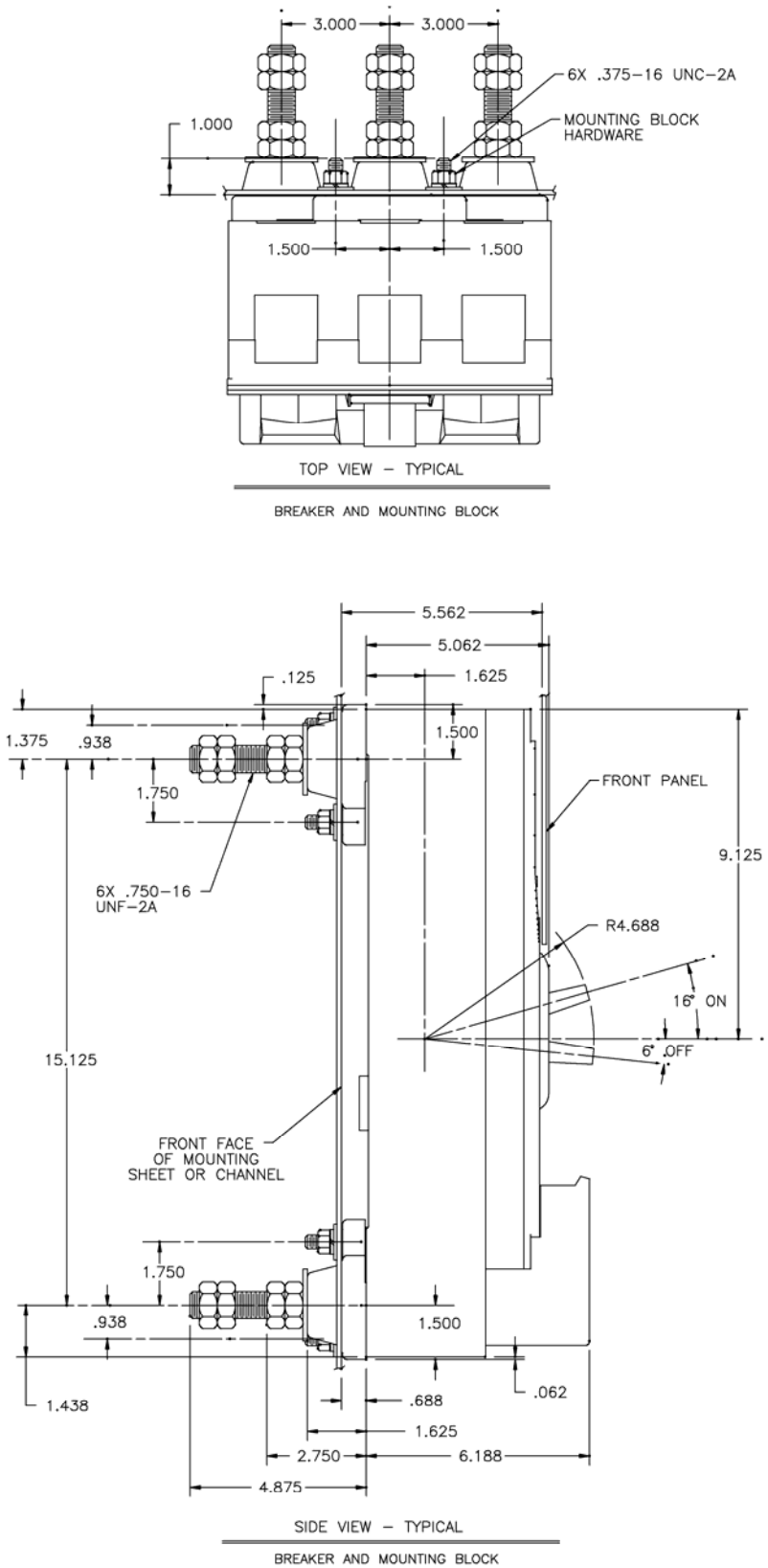


FIGURE 2. Circuit breaker types AQB-LF400 and AQB-LF402 outline and mounting block dimensions.

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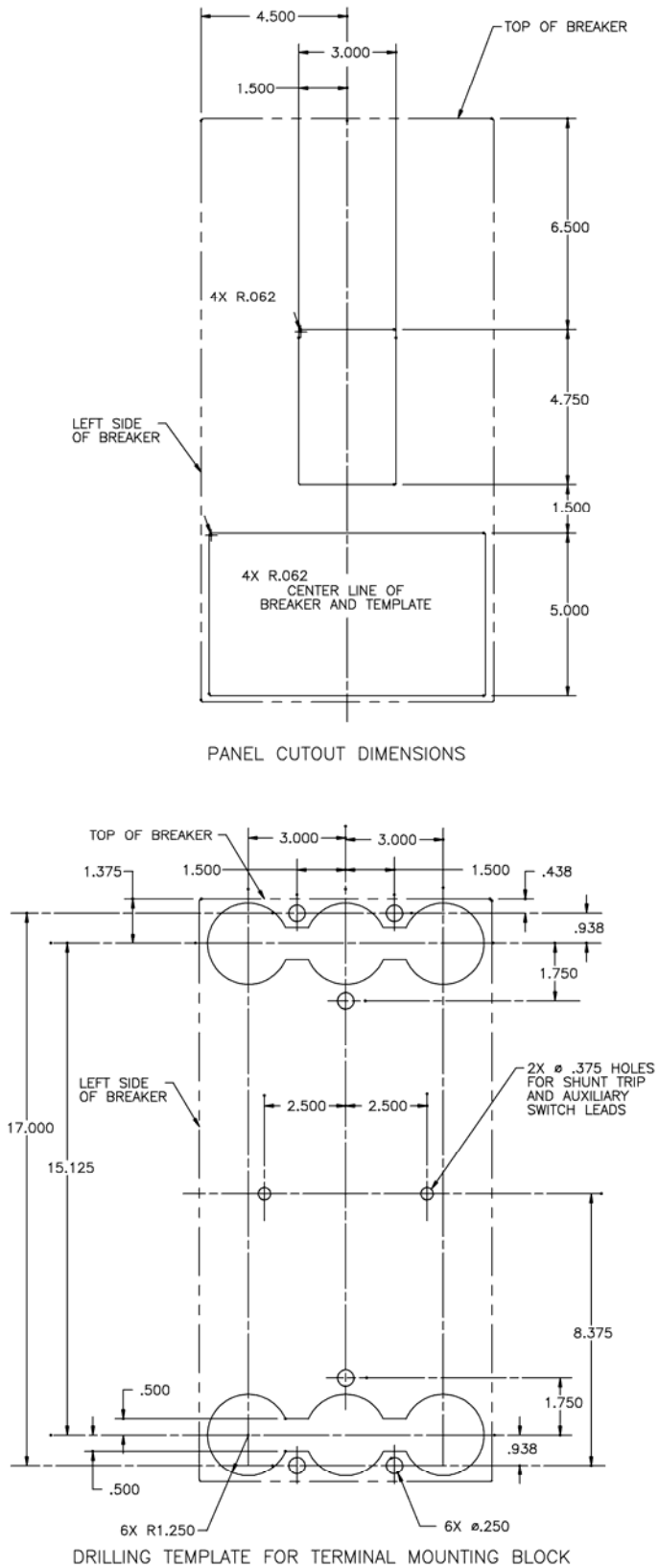


FIGURE 3. Circuit breaker types AQB-LF400 and AQB-LF402 mounting dimensions.

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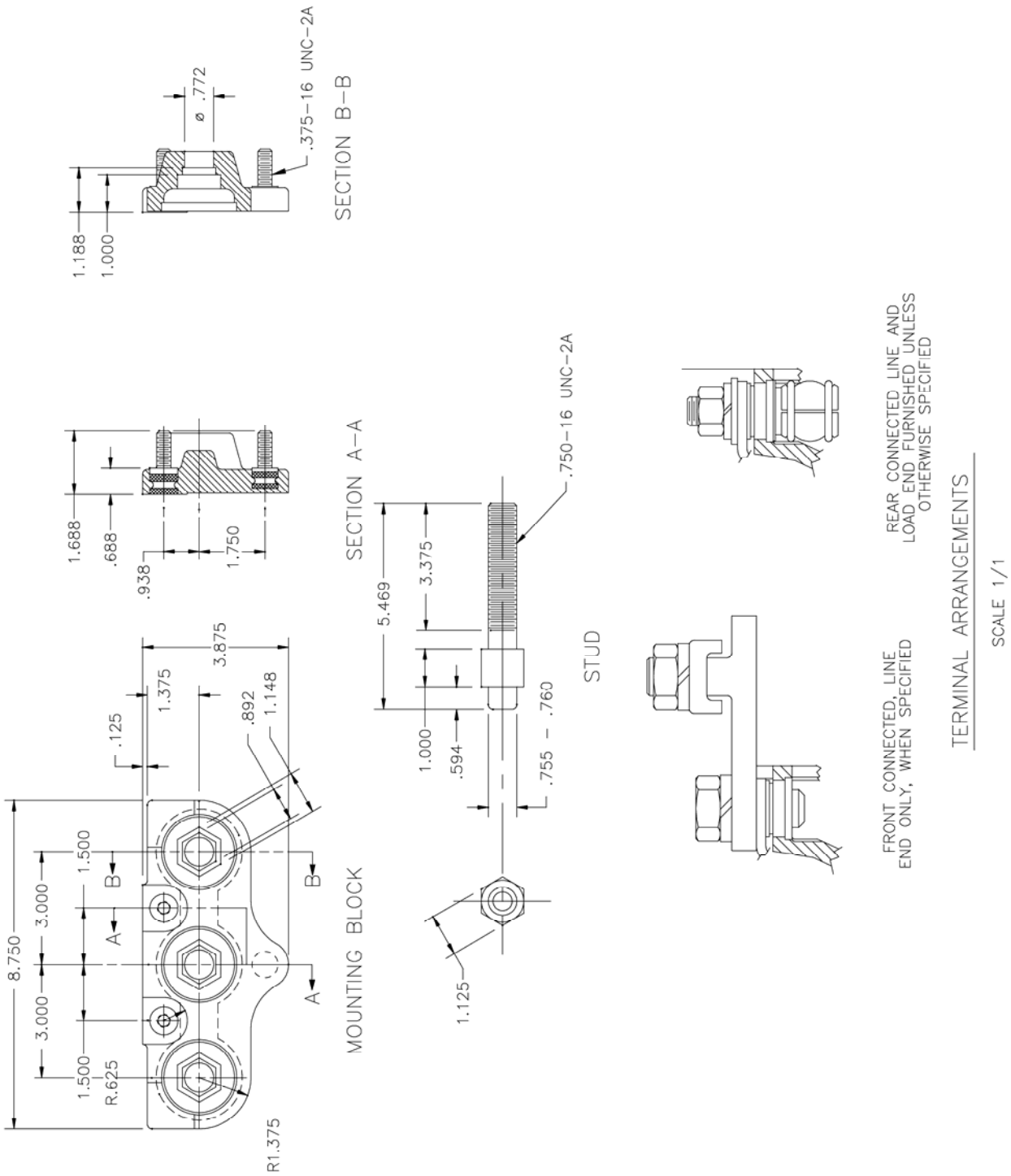
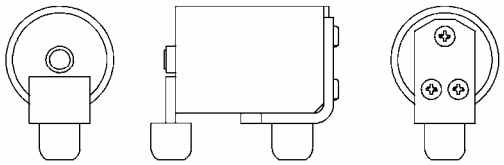
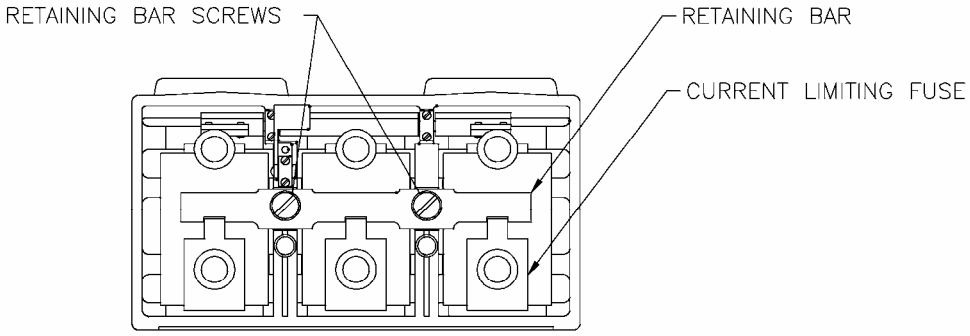


FIGURE 4. Circuit breaker types AQB-LF400 and AQB-LF402 mounting block outline dimensions.

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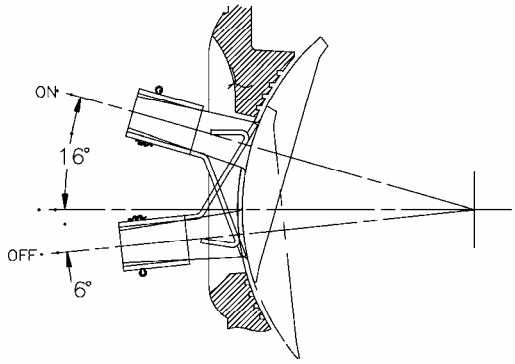
CURRENT LIMITING FUSE



CURRENT LIMITING
FUSE UNIT ASSEMBLY

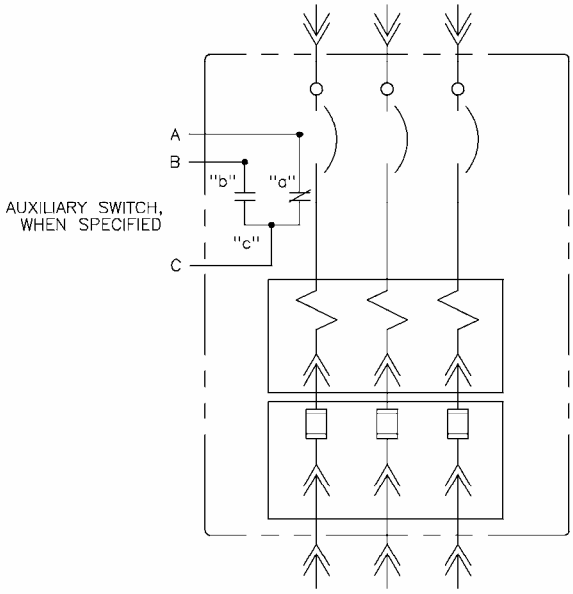
TO CHANGE CURRENT LIMITING FUSE:

1. LOOSEN SCREWS IN THE RETAINING BAR AND REMOVE THE BAR.
2. REMOVE BLOWN FUSE OR FUSES (BLOWN FUSE INDICATED BY EXTENDED FUSE PLUNGER BUTTON).
3. ASSEMBLE IN REVERSE ORDER.



CIRCUIT BREAKER
HANDLE LOCKING DEVICE

(NOT SUPPLIED WITH BREAKER)



TYPICAL WIRING DIAGRAM
(BREAKER CLOSED)

FIGURE 5. Circuit breaker types AQB-LF400 and AQB-LF402 typical wiring diagram, handle locking device, and current limiting fuse unit assembly.

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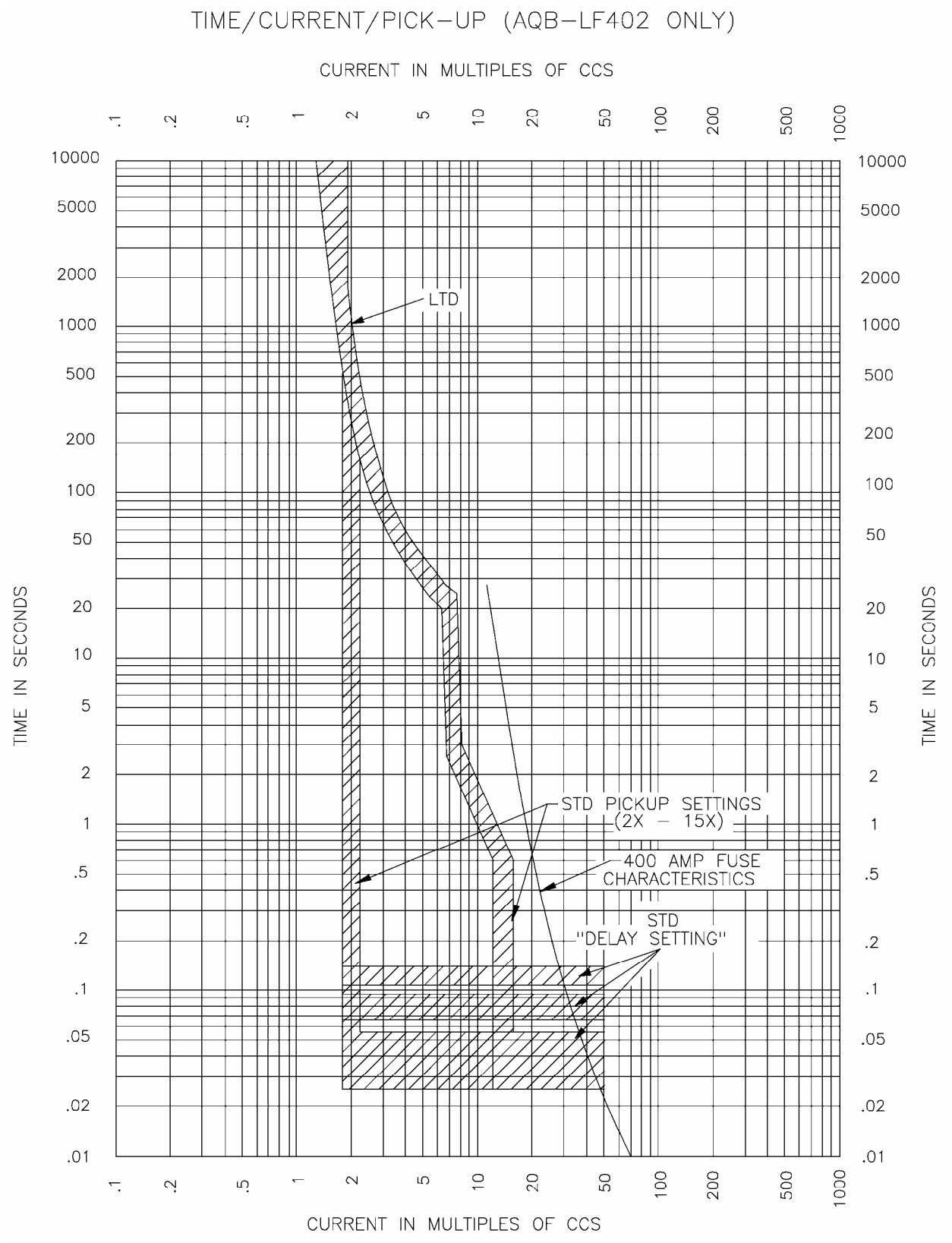


FIGURE 6. Circuit breaker type AQB-A402 time-current response.

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing Activity:
Navy – SH
(Project 5925-N185-011)

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